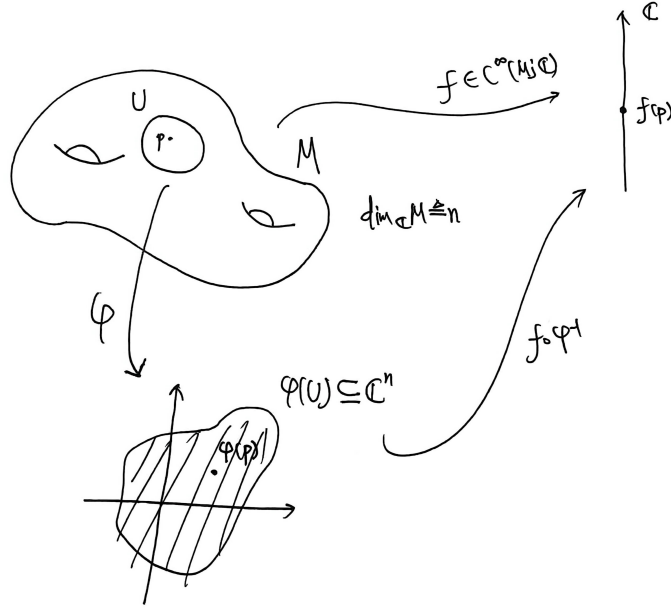


Holomorphic Jacobian

Let M be a complex manifold and $\dim_{\mathbb{C}} M = n$, let $(U, \varphi) = (U; z^1, z^2, \dots, z^n)$ be a local coordinate for $U \subseteq M$. Under the canonical identification $\mathbb{C}^n \cong \mathbb{R}^{2n}$ via $z^j \triangleq x^j + \sqrt{-1}y^j$ for $1 \leq j \leq n$, we can view $\varphi : U \rightarrow \mathbb{C}^n$ as a coordinate map $\varphi : U \rightarrow \mathbb{R}^{2n}$ with local coordinate $(x^1, y^1, x^2, y^2, \dots, x^n, y^n)$, hence we get the local frame $\{\frac{\partial}{\partial x^j}, \frac{\partial}{\partial y^j}\}_{j=1}^n$ for TU . They act on $C^\infty(U; \mathbb{C})$



by

$$\frac{\partial}{\partial x^j} \Big|_p f := \frac{\partial}{\partial x^j} \Big|_{\varphi(p)} (f \circ \varphi^{-1}), \quad \frac{\partial}{\partial y^j} \Big|_p f := \frac{\partial}{\partial y^j} \Big|_{\varphi(p)} (f \circ \varphi^{-1}). \quad (1 \leq j \leq n, \forall f \in C_p^\infty(U; \mathbb{C})).$$

We define

$$\frac{\partial}{\partial z^j} := \frac{1}{2} \left(\frac{\partial}{\partial x^j} - \sqrt{-1} \frac{\partial}{\partial y^j} \right), \quad \frac{\partial}{\partial \bar{z}^j} := \frac{1}{2} \left(\frac{\partial}{\partial x^j} + \sqrt{-1} \frac{\partial}{\partial y^j} \right)$$

just like what we do in the theory of complex analysis.

Lemma 1. Let M be a complex manifold such that $\dim_{\mathbb{C}} M = n$ and $f \in C^\infty(U; \mathbb{C})$ on a subset U . Suppose $(U; z^1, z^2, \dots, z^n)$ is a local holomorphic coordinate, then

$$f \in \mathcal{O}(U) \iff \frac{\partial f}{\partial \bar{z}^j} \equiv 0$$

for $\forall 1 \leq j \leq n$.

Proof: For $\forall p \in U$, we have

$$\frac{\partial f}{\partial \bar{z}^j}(p) = \frac{\partial (f \circ \varphi^{-1})}{\partial \bar{z}^j}(\varphi(p)) = 0,$$

where the last equality holds by the holomorphy of $f \circ \varphi^{-1} \in \mathcal{O}(\hat{U})$, $\hat{U} := \varphi(U) \subseteq \mathbb{C}^n$. ■

Recall: Recall that for smooth manifolds M and N , the differential of $F : M \rightarrow N$ at the point $p \in M$ is the linear map $dF_p : T_p M \rightarrow T_{F(p)} N$, which yields a bundle homomorphism

$dF \in \text{Hom}_{\mathbb{R}}(TM; TN)$. The complexification of $dF_p \in \text{Hom}_{\mathbb{R}}(T_p M; T_{F(p)} N)$ is the complex-linear map $(dF_p)_{\mathbb{C}} \in \text{Hom}_{\mathbb{C}}((T_p M)_{\mathbb{C}}; (T_{F(p)} N)_{\mathbb{C}})$ (we do this by complex extension of dF_p). However, we will denote the associated bundle homomorphism $(dF)_{\mathbb{C}}$ by $DF : (TM)_{\mathbb{C}} \rightarrow (TN)_{\mathbb{C}}$.

• Before we give the local coordinate representation of the map $DF(p) \in \text{Hom}_{\mathbb{C}}((T_p M)_{\mathbb{C}}; (T_{F(p)} N)_{\mathbb{C}})$, we consider a simpler case as follows: Let $f : \mathbb{C} \rightarrow \mathbb{C}^2$ be a smooth complex map, so we can write

$$f : \mathbb{C} \rightarrow \mathbb{C}^2$$

$$z \mapsto f(z) = (f^1(z), f^2(z)),$$

so if we write

$$f^1 \triangleq U^1 + \sqrt{-1}V^1, \quad f^2 \triangleq U^2 + \sqrt{-1}V^2. \quad U^i, V^i \in C^\infty(\mathbb{R}^2; \mathbb{R}) \text{ for } i = 1, 2.$$

i.e., for $\forall z \triangleq x + \sqrt{-1}y$ with $x, y \in \mathbb{R}$,

$$f^1(z) = U^1(x, y) + \sqrt{-1}V^1(x, y), \quad f^2(z) = U^2(x, y) + \sqrt{-1}V^2(x, y),$$

which means

$$f(z) = (f^1(z), f^2(z)) = (U^1(x, y) + \sqrt{-1}V^1(x, y), U^2(x, y) + \sqrt{-1}V^2(x, y)).$$

So we can express $f : \mathbb{C} \rightarrow \mathbb{C}^2$ as

$$f(z) \triangleq U(x, y) + \sqrt{-1}V(x, y),$$

where

$$U(x, y) = (U^1(x, y), U^2(x, y)), \quad V(x, y) = (V^1(x, y), V^2(x, y))$$

for $\forall z = x + \sqrt{-1}y \in \mathbb{C}$. More over, $f : \mathbb{C} \rightarrow \mathbb{C}^2$ can also be viewed as

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}^4$$

$$(x, y) \mapsto (U^1(x, y), V^1(x, y), U^2(x, y), V^2(x, y))$$

hence if we write (u^1, v^1, u^2, v^2) as the standard coordinate for $\mathbb{R}^4$, then the real differential df_p at $p \in \mathbb{R}^2$ is given by

$$\begin{aligned} df_p \left(\frac{\partial}{\partial x} \Big|_p \right) &= \frac{\partial U^1}{\partial x}(p) \frac{\partial}{\partial u^1} \Big|_{f(p)} + \frac{\partial V^1}{\partial x}(p) \frac{\partial}{\partial v^1} \Big|_{f(p)} + \frac{\partial U^2}{\partial x}(p) \frac{\partial}{\partial u^2} \Big|_{f(p)} + \frac{\partial V^2}{\partial x}(p) \frac{\partial}{\partial v^2} \Big|_{f(p)} \\ &= \sum_{k=1}^2 \frac{\partial U^k}{\partial x}(p) \frac{\partial}{\partial u^k} \Big|_{f(p)} + \sum_{k=1}^2 \frac{\partial V^k}{\partial x}(p) \frac{\partial}{\partial v^k} \Big|_{f(p)}. \end{aligned}$$

So if we generalize the discussion above to the case that $f : \mathbb{C}^m \rightarrow \mathbb{C}^n$, then we can see that $df_p : T_p \mathbb{R}^{2m} \rightarrow T_{f(p)} \mathbb{R}^{2n}$ is given by

$$\begin{aligned} df_p \left(\frac{\partial}{\partial x^j} \Big|_p \right) &= \sum_{k=1}^n \frac{\partial U^k}{\partial x^j}(p) \frac{\partial}{\partial u^k} \Big|_{f(p)} + \sum_{k=1}^n \frac{\partial V^k}{\partial x^j}(p) \frac{\partial}{\partial v^k} \Big|_{f(p)}, \\ df_p \left(\frac{\partial}{\partial y^j} \Big|_p \right) &= \sum_{k=1}^n \frac{\partial U^k}{\partial y^j}(p) \frac{\partial}{\partial u^k} \Big|_{f(p)} + \sum_{k=1}^n \frac{\partial V^k}{\partial y^j}(p) \frac{\partial}{\partial v^k} \Big|_{f(p)} \end{aligned}$$

for all $1 \leq j \leq m$.

Proposition 1. (Local Holomorphic coordinate representation for $DF(p)$)

Let M, N be two complex manifolds such that $\dim_{\mathbb{C}} M = m, \dim_{\mathbb{C}} N = n$. Let $F : M \rightarrow N$ be a smooth map. Then for any $p \in M$ and the local coordinate $(U; z^1, \dots, z^m)$ of $p \in M$ and $(V; w^1, \dots, w^n)$ of $F(p) \in N$, we have

$$\begin{aligned} DF(p) \left(\frac{\partial}{\partial z^j} \Big|_p \right) &= \sum_{k=1}^n \frac{\partial F^k}{\partial z^j}(p) \frac{\partial}{\partial w^k} \Big|_{F(p)} + \sum_{k=1}^n \frac{\partial \overline{F^k}}{\partial z^j}(p) \frac{\partial}{\partial \overline{w^k}} \Big|_{F(p)}, \\ DF(p) \left(\frac{\partial}{\partial \overline{z^j}} \Big|_p \right) &= \sum_{k=1}^n \frac{\partial F^k}{\partial \overline{z^j}}(p) \frac{\partial}{\partial w^k} \Big|_{F(p)} + \sum_{k=1}^n \frac{\partial \overline{F^k}}{\partial \overline{z^j}}(p) \frac{\partial}{\partial \overline{w^k}} \Big|_{F(p)}. \end{aligned}$$

Proof: Express $F \triangleq U + \sqrt{-1}V$ with $U, V \in C^\infty(M; \mathbb{R})$. Thus we see that

$$F^k = U^k + \sqrt{-1}V^k, \quad 1 \leq k \leq m,$$

where $U^k, V^k \in C^\infty(\mathbb{R}^{2m}; \mathbb{R})$. Denote $z^j \triangleq x^j + \sqrt{-1}y^j$, $w^j \triangleq u^j + \sqrt{-1}v^j$. Thus we have the local coordinate $(U; x^1, y^1, x^2, y^2, \dots, x^n, y^n)$ of M and $(V; u^1, v^1, u^2, v^2, \dots, u^n, v^n)$ of N in the underlying real manifolds, hence

$$T_p M = \text{Span}_{\mathbb{R}} \left\{ \frac{\partial}{\partial x^j} \Big|_p, \frac{\partial}{\partial y^j} \Big|_p \right\}_{j=1}^n, \quad T_{F(p)} M = \text{Span}_{\mathbb{R}} \left\{ \frac{\partial}{\partial u^j} \Big|_{F(p)}, \frac{\partial}{\partial v^j} \Big|_{F(p)} \right\}_{j=1}^n.$$

By the theory of smooth manifolds and the discussion above, we know that

$$DF(p) : T_p M \rightarrow T_{F(p)} N$$

is given by

$$\begin{aligned} DF(p) \left(\frac{\partial}{\partial x^j} \Big|_p \right) &= \sum_{k=1}^n \frac{\partial U^k}{\partial x^j}(p) \frac{\partial}{\partial u^k} \Big|_{F(p)} + \sum_{k=1}^n \frac{\partial V^k}{\partial x^j}(p) \frac{\partial}{\partial v^k} \Big|_{F(p)}, \\ DF(p) \left(\frac{\partial}{\partial y^j} \Big|_p \right) &= \sum_{k=1}^n \frac{\partial U^k}{\partial y^j}(p) \frac{\partial}{\partial u^k} \Big|_{F(p)} + \sum_{k=1}^n \frac{\partial V^k}{\partial y^j}(p) \frac{\partial}{\partial v^k} \Big|_{F(p)}. \end{aligned}$$

By the definition, $\frac{\partial}{\partial z^j} := \frac{1}{2} \left(\frac{\partial}{\partial x^j} - \sqrt{-1} \frac{\partial}{\partial y^j} \right)$, $\frac{\partial}{\partial \overline{z^j}} := \frac{1}{2} \left(\frac{\partial}{\partial x^j} + \sqrt{-1} \frac{\partial}{\partial y^j} \right)$ and the complex-linearity of $DF(p)$, so we get

$$\begin{aligned} DF(p) \left(\frac{\partial}{\partial z^j} \right) &= DF(p) \left(\frac{1}{2} \left(\frac{\partial}{\partial x^j} - \sqrt{-1} \frac{\partial}{\partial y^j} \right) \right) \\ &= \frac{1}{2} DF(p) \left(\frac{\partial}{\partial x^j} \right) - \sqrt{-1} DF(p) \left(\frac{\partial}{\partial y^j} \right) \\ &= \sum_{k=1}^n \frac{1}{2} \left(\frac{\partial}{\partial x^j} - \sqrt{-1} \frac{\partial}{\partial y^j} \right) U^k \frac{\partial}{\partial u^k} + \sum_{k=1}^n \frac{1}{2} \left(\frac{\partial}{\partial x^j} - \sqrt{-1} \frac{\partial}{\partial y^j} \right) V^k \frac{\partial}{\partial v^k} \\ &= \sum_{k=1}^n \frac{\partial U^k}{\partial z^j}(p) \frac{\partial}{\partial u^k} \Big|_{F(p)} + \sum_{k=1}^n \frac{\partial V^k}{\partial z^j}(p) \frac{\partial}{\partial v^k} \Big|_{F(p)}, \end{aligned}$$

similarly, we have

$$DF(p) \left(\frac{\partial}{\partial \overline{z^j}} \right) = \sum_{k=1}^n \frac{\partial U^k}{\partial \overline{z^j}}(p) \frac{\partial}{\partial u^k} \Big|_{F(p)} + \sum_{k=1}^n \frac{\partial V^k}{\partial \overline{z^j}}(p) \frac{\partial}{\partial v^k} \Big|_{F(p)}.$$

Note that we have

$$\frac{\partial}{\partial u^k} = \frac{\partial}{\partial w^k} + \frac{\partial}{\partial \bar{w}^k}, \quad \frac{\partial}{\partial v^k} = \sqrt{-1} \left(\frac{\partial}{\partial w^k} - \frac{\partial}{\partial \bar{w}^k} \right),$$

inserting these in the two equations above yields

$$DF(p) \left(\frac{\partial}{\partial z^j} \right) = \sum_{k=1}^n \frac{\partial U^k}{\partial z^j}(p) \left(\frac{\partial}{\partial w^k} + \frac{\partial}{\partial \bar{w}^k} \right) \Big|_{F(p)} + \sum_{k=1}^n \frac{\partial V^k}{\partial z^j}(p) \sqrt{-1} \left(\frac{\partial}{\partial w^k} - \frac{\partial}{\partial \bar{w}^k} \right) \Big|_{F(p)},$$

i.e., we have

$$\begin{aligned} DF(p) \left(\frac{\partial}{\partial z^j} \right) &= \sum_{k=1}^n \frac{\partial(U^k + \sqrt{-1}V^k)}{\partial z^j}(p) \frac{\partial}{\partial w^k} \Big|_{F(p)} + \sum_{k=1}^n \frac{\partial(U^k - \sqrt{-1}V^k)}{\partial z^j}(p) \frac{\partial}{\partial \bar{w}^k} \Big|_{F(p)} \\ &= \sum_{k=1}^n \frac{\partial F^k}{\partial z^j}(p) \frac{\partial}{\partial w^k} \Big|_{F(p)} + \sum_{k=1}^n \frac{\partial \bar{F}^k}{\partial z^j}(p) \frac{\partial}{\partial \bar{w}^k} \Big|_{F(p)} \end{aligned}$$

and similarly we have

$$DF(p) \left(\frac{\partial}{\partial \bar{z}^j} \right) = \sum_{k=1}^n \frac{\partial F^k}{\partial \bar{z}^j}(p) \frac{\partial}{\partial w^k} \Big|_{F(p)} + \sum_{k=1}^n \frac{\partial \bar{F}^k}{\partial \bar{z}^j}(p) \frac{\partial}{\partial \bar{w}^k} \Big|_{F(p)},$$

which completes the proof. ■

Remark: By this proposition, we see that for the smooth map $F : M^m \longrightarrow N^n$, if the local holomorphic coordinates $(U; z^1, \dots, z^m)$ and $(V; w^1, \dots, w^n)$ are given such that $p \in U$ and $F(p) \in V$, then we have

$$\begin{aligned} DF(p) \left(\frac{\partial}{\partial z^1} \quad \dots \quad \frac{\partial}{\partial z^m} \quad \frac{\partial}{\partial \bar{z}^1} \quad \dots \quad \frac{\partial}{\partial \bar{z}^m} \right) \\ = \left(\frac{\partial}{\partial w^1} \quad \dots \quad \frac{\partial}{\partial w^n} \quad \frac{\partial}{\partial \bar{w}^1} \quad \dots \quad \frac{\partial}{\partial \bar{w}^n} \right) \begin{pmatrix} \left(\frac{\partial F^i}{\partial z^j}(p) \right) & \left(\frac{\partial \bar{F}^i}{\partial z^j}(p) \right) \\ \left(\frac{\partial F^i}{\partial \bar{z}^j}(p) \right) & \left(\frac{\partial \bar{F}^i}{\partial \bar{z}^j}(p) \right) \end{pmatrix}. \end{aligned}$$

Definition 1.(Holomorphic Jacobian Matrix of a map)

Let M, N be two complex manifolds and let $F : M \longrightarrow N$ be a smooth map, we denote

$$J^{\text{Hol}} F := \left(\frac{\partial F^i}{\partial z^j} \right) \in M_{\dim_{\mathbb{C}} N \times \dim_{\mathbb{C}} M}(\mathbb{C}),$$

which is called the **holomorphic Jacobian of F** .

If $F : M \longrightarrow N$ is a holomorphic map, i.e., $F \in \mathcal{O}(M; N)$, then we see that representation matrix of F under the local frame $\left\{ \frac{\partial}{\partial z^j}, \frac{\partial}{\partial \bar{z}^j} \right\}_{j=1}^{\dim_{\mathbb{C}} M}$ and $\left\{ \frac{\partial}{\partial w^j}, \frac{\partial}{\partial \bar{w}^j} \right\}_{j=1}^{\dim_{\mathbb{C}} N}$ is given as follows

$$F \in \mathcal{O}(M; N) \rightsquigarrow DF(p) = \begin{pmatrix} J^{\text{Hol}} F(p) & O \\ O & J^{\text{Hol}} F(p) \end{pmatrix}.$$

- Under the canonical isomorphism $\mathbb{C}^n \cong \mathbb{R}^{2n}$, since $\mathbb{R}^{2n}$ has the orientation form

$$\omega_{\mathbb{R}^{2n}} = dx^1 \wedge dy^1 \wedge \dots \wedge dx^n \wedge dy^n,$$

for $z^j = x^j + \sqrt{-1}y^j$, we get

$$dx^j = \frac{1}{2}(dz^j + d\bar{z}^j), \quad dy^j = \frac{1}{2\sqrt{-1}}(dz^j - d\bar{z}^j),$$

which implies that

$$dx^j \wedge dy^j = \frac{\sqrt{-1}}{2} dz^j \wedge d\bar{z}^j \quad \text{for } 1 \leq j \leq n.$$

Thus we get the canonical orientation form $\omega_{\mathbb{C}^n} \in \Omega_{\mathbb{C}}^n(\mathbb{C}^n)$, which is

$$\omega_{\mathbb{C}^n} = \left(\frac{\sqrt{-1}}{2} \right)^n dz^1 \wedge d\bar{z}^1 \wedge dz^2 \wedge d\bar{z}^2 \wedge \cdots \wedge dz^n \wedge d\bar{z}^n.$$

So if the map $F : M \rightarrow M$ is a local biholomorphism, then for any local coordinate $(U; z^1, \dots, z^n)$, we get

$$\begin{aligned} F^*(dz^1 \wedge \cdots \wedge dz^n \wedge d\bar{z}^1 \wedge \cdots \wedge d\bar{z}^n) &= \det \begin{pmatrix} J^{\text{Hol}} F & O \\ O & \overline{J^{\text{Hol}} F} \end{pmatrix} dz^1 \wedge \cdots \wedge dz^n \wedge d\bar{z}^1 \wedge \cdots \wedge d\bar{z}^n \\ &= |\det J^{\text{Hol}} F|^2 dz^1 \wedge \cdots \wedge dz^n \wedge d\bar{z}^1 \wedge \cdots \wedge d\bar{z}^n \end{aligned}$$

by computation, we can know

$$\omega_{\mathbb{C}^n} = (-1)^{\frac{(n-1)n}{2}} \left(\frac{\sqrt{-1}}{2} \right)^n dz^1 \wedge \cdots \wedge dz^n \wedge d\bar{z}^1 \wedge \cdots \wedge d\bar{z}^n,$$

hence on the local holomorphic coordinate $(U; z^1, z^2, \dots, z^n)$, we get

$$F^* \omega_{\mathbb{C}^n} = |\det J^{\text{Hol}} F|^2 \omega_{\mathbb{C}^n}, \quad \text{where } |\det J^{\text{Hol}} F(p)|^2 > 0 \text{ for all } p,$$

since M is locally $\mathbb{C}^n$, from this we can see that the local biholomorphism $F : M \rightarrow M$ is orientation-preservation.